Password Strength Analyzer with Custom Wordlist Generator

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1. Introduction

In today's digital world, weak passwords are one of the major causes of cyber attacks such as brute force and dictionary attacks. This project aims to analyze password strength using entropy calculation and pattern detection while also generating custom wordlists to simulate real-world password guessing attacks.

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2. Abstract

The Password Strength Analyzer evaluates user-entered passwords based on length, complexity, entropy, and common password databases. It categorizes passwords as weak, medium, or strong. Additionally, the tool generates a customized wordlist based on user-provided personal information such as name and birth year to demonstrate how attackers exploit predictable password patterns.

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3. Tools Used

- Python
- Math Library (Entropy Calculation)
- Tkinter
- Optional: zxcvbn Library
- re (regex)

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4. Steps Involved

1. User inputs password
2. System checks length and complexity
3. Entropy value calculated
4. Password checked against common passwords
5. Strength score generated

6. Custom wordlist generated
7. Wordlist exported in .txt format

Testing:
The tool was tested using weak, medium, and strong password combinations such as:

- 12345 → Weak
- muskan123 → Medium
- Muskan@123 → Strong

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5. Conclusion

This project demonstrates how password strength can be evaluated using entropy and pattern detection. It highlights the risks of predictable passwords and educates users about creating secure passwords. The custom wordlist feature also helps understand real-world password attack techniques.